



Full Stack Development Week 07

Topics covered:

- `useReducer`
- `useMemo`
- Accessing your Cloud SQL database

Activity 1: `useReducer`

As discussed in last weeks lecture, the `useReducer` hook is a powerful hook in React that's particularly useful when you have complex state logic.

Why use `useReducer`?

The `useReducer` hook is beneficial when dealing with complex state logic involving multiple fields, provides predictable state updates through a reducer function, and helps centralize state management logic.

Practical Example: Shopping Cart

Let's create a simple shopping cart using `useReducer`.

We should have 3 actions:

- `ADD_ITEM`
- `REMOVE_ITEM`
- `CLEAR_CART`

Exercise

Review the activity code and modify it to:

- Add a quantity field to each item
- Implement an "UPDATE_QUANTITY" action
- Display the quantity against the items

To discuss

- Why is `useReducer` better than `useState` for this shopping cart example?
- How would you handle async operations (like fetching product data) with `useReducer`?

Activity 2: `useMemo`

The `useMemo` hook is used for performance optimization by memoizing expensive computations. It only recomputes the memoized value when one of the dependencies has changed.

Why use `useMemo`?

- Prevents unnecessary recalculations of expensive computations
- Optimizes performance for "expensive" (in terms of time) functions

Exercise

Let's review a simple activity where we have a very large list of items (10,000+) and we need to search them, plus we also have a counter on the screen as well. Let's see how `useMemo` makes a difference by creating two examples

To discuss:

- What happens if we increase the counters value, will all 10,000 items rerender if we are not using `useMemo`
- What's the performance impact of removing `useMemo` in this example?
- Are there cases where using `useMemo` might actually hurt performance?

Activity 3: Access your Cloud SQL database

Assignment 2 will require you to save web application data in database tables instead of `localStorage`. For this course, we will use MySQL database hosted in Cloud- this is so that you DO NOT HAVE to install it on your machine. While we will start looking at code interacting with the database using ORM from next week, it is important that all of you get accustomed with your MySQL database account and logistics around it.

[!WARNING] Do not install or run MySQL locally on your machine. We will be using the provided Cloud SQL database for this course. Running MySQL locally could lead to inconsistencies between your local and cloud environments, potentially causing issues with assignments and making it harder to debug problems. Stick to using the provided Cloud SQL instance to ensure consistency across development environments.

Login Details

IP: 209.38.26.237 Username: Your student number with a capital S Password: Same as username initially

phpMyAdmin URL: <https://getmysql.com/phpmyadmin/>

Note: If you are unable to login using the above, please contact matt.hayward@rmit.edu.au

Note: Please do not contact ITS for access, this database is maintained by us for this course only.

How to access via phpMyAdmin

phpMyAdmin is a free web-based administration tool for MySQL databases. It provides an intuitive graphical interface that allows you to:

- View, create, edit, and delete databases and tables

- Execute SQL queries directly through the web interface
- Import and export data in various formats (SQL, CSV, XML, etc.)

To access phpMyAdmin:

1. Navigate to <https://getmysql.com/phpmyadmin/> in your web browser
2. Enter your student number (with capital S) as both username and password
3. After logging in, you'll see your databases listed in the left sidebar
4. Click on a database name (your database is the same as your username) to explore its tables and structure

For security reasons:

- Change your password after first login

Changing your password

Run the following query using the SQL tab to change your password.

```
ALTER USER 'S123456' IDENTIFIED BY 'newpasswordhere'
```

Then run this query

```
FLUSH PRIVILEGES
```

Creating some tables

We will now load some sample data into our database. Download and open the [activity3.sql](#) file from this weeks code examples and execute it via the SQL tab (by coping all of the contents into the SQL tab and clicking "Go" in the bottom right)

Once this is done, refresh the page and you should now see three tables:

- customers
- orders
- order_items

Explore these three tables, seeing if you can write a query that pulls a list of items ordered by [Alice](#) across all of their orders.

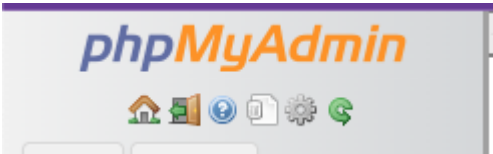
Creating ERDs (Entity Relationship Diagrams)

You can also use phpMyAdmin to create nice ERDs of your database structure.

Click on your database in the left hand of the page and then click "Designer"

Logging out

To logout, click the little icon with the green arrow leaving the door in the top left of the screen.



What's next?

In future weeks, we'll connect our applications to our MySQL databases, replacing our local storage from Assignment 1 with an actual database.

